



# Natural language processing course project report

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## Abstract

We are developing an AI assistant for the UL FRI student office that answers administrative questions in Slovenian and English from official faculty documents. We implemented an end-to-end pipeline that crawls UL/FRI web content, parses HTML, PDF, and DOCX into structured text, applies section-aware chunking, embeds the chunks with a multilingual sentence-transformer, indexes them with FAISS, and supports dense, BM25+dense hybrid, and cross-encoder reranked retrieval feeding an instruction-tuned generator. We evaluate the pipeline with both classical retrieval metrics (Recall@k, MRR@k, nDCG@k) and a RAGAS-style LLM-as-judge protocol scoring faithfulness, hallucination, refusal, and overall quality. To make the LLM judgments meaningful we first audit the human-written reference answers against the indexed corpus, producing a cleaned evaluation set. On that set, an embedder  $\times$  retrieval-mode sweep followed by a chunk-size sweep identifies `multilingual-e5-base` with hybrid retrieval, cross-encoder reranking, top- $k = 4$ , and 400-token chunks as the best configuration, with a Sonnet-graded faithfulness of 0.740 and 11/11 correct refusals on out-of-corpus questions.

## Keywords

Retrieval augmented generation, dense retrieval, hybrid retrieval, multilingual embeddings, student affairs chatbot

## Introduction

The purpose of this project is to develop an AI assistant for the student office of the Faculty of Computer and Information Science of the University of Ljubljana (UL FRI). The system serves as a chatbot that answers students' administrative questions about study programmes, study regulations, and the work of the student office, in Slovenian and English. We base the assistant on Retrieval Augmented Generation [1], a paradigm that has been shown to fit university-style FAQ chatbots [2, 3] and that, in recent surveys of educational deployments, addresses key issues such as hallucination and outdated knowledge [4]. Our pipeline design takes cues from prior RAG-for-admissions and university-resources work that compares dense, hybrid, and reranked retrieval against varied chunking strategies [5, 6], and from studies on local LLMs for prospective-student chatbots [7] and on RAG vs. fine-tuning for knowledge-intensive tasks [8].

## Methods

### Pipeline overview

The implementation is a complete data-to-index-to-answer pipeline organized as runs: each run stores its raw downloads, parsed JSONL documents, chunks, and the FAISS index in its own directory.

### Data collection and parsing

A lightweight crawler starts from a curated seed list and performs breadth-first traversal with separate depth limits for `fri.uni-lj.si`, other `*.uni-lj.si` domains, and external domains. It downloads HTML pages and linked PDF/DOCX attachments, complemented by sitemaps and RSS/Atom feeds, respects `robots.txt`, throttles requests, and stores SHA-256 hashes plus HTTP timestamps in a per-run `manifest.jsonl`. Update crawls reuse `ETag/If-Modified-Since`. Documents are parsed into a common JSON structure with a stable URL-derived `doc_id`, title, language, and ordered named sections: HTML via BeautifulSoup (with optional `trafilatura`), PDF via DocumentCloud with Tesseract OCR (or PyMuPDF as a fallback), and DOCX via `python-docx`. Image-only sources are indexed via required sidecar text descriptions.

### Chunking, embeddings and retrieval

Documents are split with a section-aware sliding window measured in whitespace tokens (default 400 tokens, overlap 80). Each chunk is contextualized with the document title and section name, gets a stable `chunk_id`, and is deduplicated with a SimHash filter. We embed chunks with `intfloat/multilingual-e5-base`, applying E5-style `passage:query:` prefixes and L2 normaliza-

tion, and index with FAISS IndexFlatIP (equivalent to cosine under normalization). Retrieval supports three modes: dense (top- $k$  from FAISS), hybrid (BM25 fused with dense scores after min-max normalization on the union of both top lists), and hybrid + cross-encoder rerank (a sentence-transformers CrossEncoder reranks a candidate pool of size 20). A lightweight additive recency boost (default weight 0.08, half-life 365 days) and a `retrieval_weak` score-threshold flag are applied on top, the latter both as a UI signal and as a prompt-side warning. For end-to-end QA, retrieved chunks are formatted into a numbered context block, and a language-specific system prompt instructs the generator to answer *only* from that context and in the detected language.

### Evaluation set and reference audit

We evaluate on a manually written JSONL question set. Each entry carries the expected language and expected keywords, and positives also carry `relevant_chunk_ids` for chunk-level retrieval metrics, plus a free-form `reference_answer` used as the human-curated gold for the LLM-as-judge protocol below; the set contains 61 questions (51 positive, 10 negative). To prevent low scores caused by reference-vs.-corpus drift rather than retrieval errors, we run a Sonnet-graded fact-check before any cross-configuration comparison: for each positive question, we retrieve a top-30 BM25  $\cup$  dense pool from the current index and ask Claude Sonnet (`claude-sonnet-4-6`,  $T=0$ ) to label every claim in the reference as `supported`, `partial`, or `unsupported`, and to optionally produce a grounded rewrite. The fixed application policy is: `supported` keeps the original, `partial/unsupported` with a rewrite uses the rewrite (flagged `auto_rewritten`), and `unsupported` without a rewrite is dropped; negatives are never modified. The cleaned set is written as a versioned artefact `questions_full_v2.json`.

### LLM-as-judge protocol

We adopt a RAGAS-style protocol: for every row we send the question, the retrieved top- $k$  chunks, the model answer, and the reference answer to Claude with a strict-JSON rubric covering *faithfulness*, *answer relevance*, and *context relevance* (each in  $[0,1]$ ), binary *hallucination* and *refusal* flags, and an integer *overall* score in  $[1,5]$ . Negatives are scored under the same rubric (a correct refusal is rewarded with high faithfulness). All judge calls use  $T=0$ . To control cost without sacrificing the bottleneck quality, we use Claude Sonnet (`claude-sonnet-4-6`) for the reference audit and the headline re-judgment, and Claude Haiku (`claude-haiku-4-5`) for the comparison sweeps; the two judges agreed to within 0.001 in in-scope faithfulness on the winning configuration (0.739 vs. 0.740), confirming that the cheaper Haiku sweep is a faithful proxy for ranking decisions. Per-configuration indices are built by short driver scripts that symlink `raw/` from the source crawl and pass `EMBEDDING_MODEL/CHUNK_SIZE/CHUNK_OVERLAP` as environment overrides; the chosen parameters are persisted

to `embedding-info.json` alongside the index and rechecked on load to prevent silent embedder/chunk-size mismatches.

## Results

### Corpus, evaluation set, and reference audit

The full crawl yields 2,742 parsed documents and 29,296 chunks at the default 400/80 chunk policy, up from 176 documents on the initial *Test* corpus; disabling OCR (PyMuPDF text extraction only) for the full crawl reduced index-build wall time from  $\sim 4$  hours to  $\sim 14$  minutes for  $\sim 500$  PDFs without measurable retrieval-quality loss. Running the Sonnet audit on the original references against the full-crawl index revealed that only 9/47 ( $\sim 19\%$ ) of the in-scope references were fully supported as written; 31 were classified `partial` and rewritten using only chunk-attested facts, 7 were `unsupported` (3 dropped, 4 rewritten where Sonnet found alternative chunks), and 14 were skipped (no factual reference, or negative). The cleaned `questions_full_v2.json` contains 54 in-scope and 11 negative questions and is the basis for everything that follows.

### Chunk-level retrieval (Test corpus)

On the Test corpus with chunk-level annotations, dense retrieval at  $k=4$  already reaches 88.0% Recall@4 and 66.3% nDCG@4; adding a cross-encoder reranker lifts these to 100.0% and 93.4% respectively, while hybrid + rerank is essentially tied (100.0% / 93.9%). A retrieval-only chunk-size sweep at  $k=6$  on the Test corpus shows that sizes 200–500 all reach  $\sim 97.9\%$  Recall@6, with 300 tokens giving the highest precision; the LLM-judged sweep on the full crawl below revisits this choice on a stricter metric and selects 400.

### Embedder $\times$ retrieval-mode sweep (LLM-as-judge)

Table 1 reports an embedder  $\times$  retrieval-mode sweep at top- $k=4$  on the cleaned set, judged by Claude Haiku at  $T=0$ ; refusal-side faithfulness is  $\geq 0.90$  across all six configurations. Two findings stand out. First, cross-encoder reranking is by far the dominant lever: on `e5-base`, it lifts faithfulness from 0.575 (dense) and 0.565 (hybrid) to 0.739, and context relevance from  $\sim 0.72$  to 0.857. Second, `e5-base + rerank` outperforms `e5-large + rerank` on every in-scope metric while embedding  $\sim 2.5\times$  faster: once the cross-encoder does the final ranking, the smaller first-stage embedder is sufficient.

### Chunk size on the winning configuration

Holding the winner (`e5-base`, hybrid + rerank,  $k=4$ ) fixed and varying the chunk size with overlap at 20%, the Haiku-judged in-scope faithfulness is 0.691 at 200/40 (50,546 chunks), 0.698 at 300/60 (36,331), and 0.739 at 400/80 (29,296). Larger chunks help here because the cross-encoder benefits from more local context per chunk, and 400 also yields the smallest, fastest index of the three, so we keep it as the default.

**Table 1.** Embedder × retrieval-mode sweep on `questions_full_v2.json` (Haiku judge, top- $k=4$ ,  $T=0$ ).

| Embedder                    | Mode                   | Faithfulness | Hallucination | Ctx rel.     | Overall (1–5) |
|-----------------------------|------------------------|--------------|---------------|--------------|---------------|
| multilingual-e5-base        | dense                  | 0.575        | 59.1%         | 0.725        | 3.000         |
| multilingual-e5-base        | hybrid                 | 0.565        | 63.6%         | 0.715        | 2.841         |
| <b>multilingual-e5-base</b> | <b>hybrid + rerank</b> | <b>0.739</b> | <b>54.5%</b>  | <b>0.857</b> | <b>3.432</b>  |
| multilingual-e5-large       | dense                  | 0.680        | 54.5%         | 0.812        | 3.250         |
| multilingual-e5-large       | hybrid                 | 0.650        | 61.4%         | 0.787        | 3.205         |
| multilingual-e5-large       | hybrid + rerank        | 0.708        | 59.1%         | 0.866        | 3.364         |

### Headline configuration and ablation

We re-judged the winning configuration with Claude Sonnet at  $T=0$  on the full cleaned set (Table 2). The headline in-scope faithfulness is 0.740, answer relevance 0.820, context relevance 0.807, overall 3.605/5, and hallucination rate 44.2%; on the 11 negatives the system refuses correctly in 11/11 cases with refusal-side faithfulness 0.905. Decomposing the gain across stages: the original baseline (small corpus, noisy refs, dense  $k=8$ ) had Haiku-judged faithfulness of 0.34; the full crawl plus reference audit at dense  $k=4$  moves it to 0.575; hybrid alone does not help (0.565); adding cross-encoder rerank yields 0.739 (Haiku) / 0.740 (Sonnet), with hallucination dropping from 80.9% to 44.2%.

**Table 2.** Headline numbers for the winning configuration (Sonnet judge,  $T=0$ ).

| Metric             | In-scope (n=43) | Refusal (n=11) |
|--------------------|-----------------|----------------|
| Faithfulness       | <b>0.740</b>    | 0.905          |
| Answer relevance   | 0.820           | —              |
| Context relevance  | 0.807           | —              |
| Overall (1–5)      | <b>3.605</b>    | 3.818          |
| Hallucination rate | <b>44.2%</b>    | —              |
| Refused correctly  | —               | <b>11 / 11</b> |

### Generator-model evaluation (cluster, in progress)

The headline numbers above use Qwen/Qwen2.5-1.5B-Instruct, the largest generator that fits on the laptop. To compare cjvt/GaMS3-12B-Instruct, meta-llama/Llama-3.1-8B-Instruct, and mistralai/Mistral-7B-Instruct-v0.3 on the same retrieval configuration, we extended the existing Slurm wrapper to use  $k=4$  with hybrid + rerank, added a converter to the cluster’s JSONL format, and added rsync helpers and a local LLM-as-judge aggregator. A laptop smoke test on 3 questions verified the wiring; the per-generator headline table will be filled in once the cluster jobs complete.

## Discussion

The retrieval and LLM-as-judge results converge on the same conclusion: cross-encoder reranking is the single largest lever, lifting in-scope faithfulness on e5-base from 0.575

to 0.739 and saturating chunk-level Recall@ $k$  on the Test corpus at 100%. Hybrid retrieval is a wash on its own but adds value once the rerank step is on top, since the union pool gives the cross-encoder a more diverse set of candidates. Counter to expectations, multilingual-e5-base beats multilingual-e5-large once reranking is in play while embedding  $\sim 2.5\times$  faster: the cross-encoder absorbs the work that would otherwise depend on a stronger first-stage embedder. Equally important is the reference-quality finding: only  $\sim 19\%$  of the original in-scope references were fully grounded, and the original baseline’s 0.34 faithfulness was a corpus-and-references signal, not a model signal – the corpus expansion (176→2,742 docs), the Sonnet audit, and reranking together account for the entire jump to 0.740.

The main limitations are the small evaluation set (54 in-scope, 11 negatives) and the per-question variability of the LLM-as-judge: a stratified manual review against the Sonnet judge (Cohen’s  $\kappa$ ) is the next step. The headline is reported with a small Qwen generator; the per-generator comparison against GaMS3-12B, Llama-3.1-8B, and Mistral-7B is wired end-to-end and only awaits cluster runs. We also disabled OCR for full-crawl PDFs to keep build time tractable; for scanned PDFs this would silently drop content, so a future iteration should re-enable Docling/Tesseract on the subset where PyMuPDF returns near-empty text. Finally, the recency boost is a fixed exponential decay with hand-tuned half-life and would benefit from calibration against date-sensitive questions.

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